

Interpretability and Safety

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MECHANISTIC INTERPRETABILITY

We changed **one number** inside the model and it became obsessed with a bridge.

Not a prompt. Not fine-tuning. A single direction in activation space, turned up.



Golden Gate
Claude

By the end, one idea does all the work

Concepts are directions in a model's activation space. If that's true, we can *read* what a model is thinking and *write* to it.



*We build the case (Anatomy), validate it (Proof), state it (Keystone) — then exploit it (Locate + Steer) and look ahead (Frontier).
The Locate–Steer framing follows recent MI surveys (Rai et al. 2024; Zhang et al. 2026).*



Debugging

find the cause of a failure



Trust

verify what a model actually uses

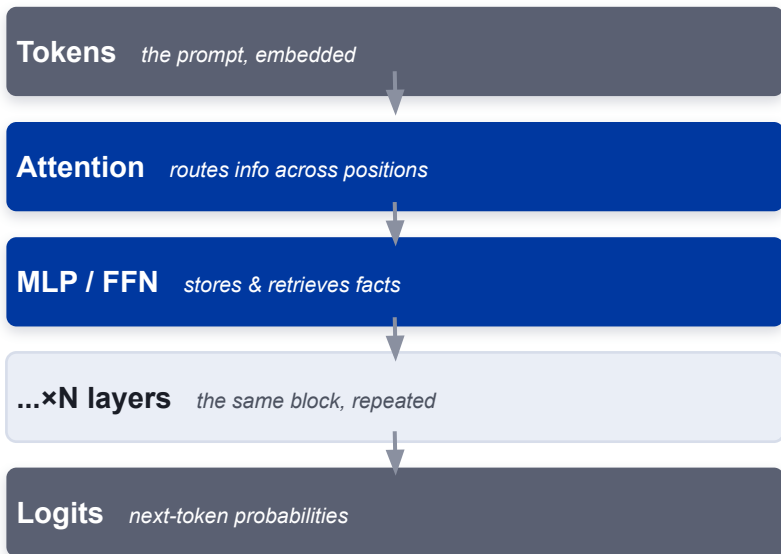


Safety

detect & steer dangerous behavior

A transformer is stacked, repeating blocks

Tokens in, logits out. Between them: a tall stack of identical blocks, each one attention followed by an MLP. We'll dissect each part in turn.



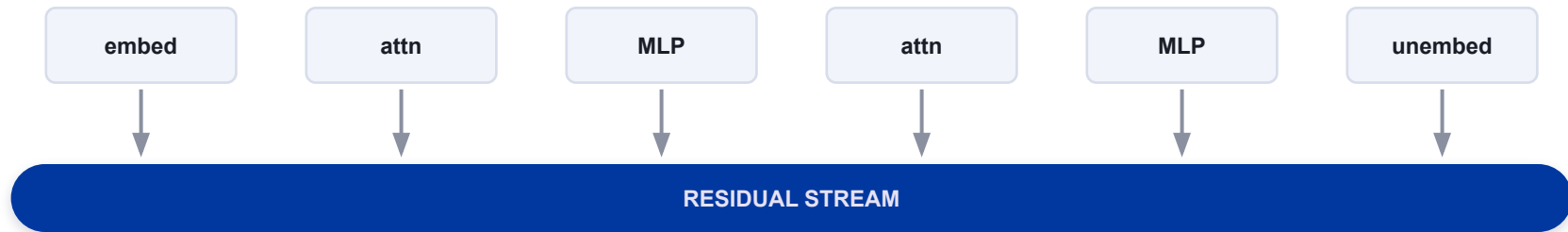
The spine: the residual stream

Every block reads from and writes to one running vector that threads the whole stack.

That shared interface is exactly what makes interpretability tools possible there's one place to look.

The residual stream, up close

A bus running through the network: each component reads the current state, adds its result back, and nothing gets overwritten contributions accumulate.



This single interface is what three core tools attach to:



Patch

swap activations to test cause



Steer

add a direction to push behavior

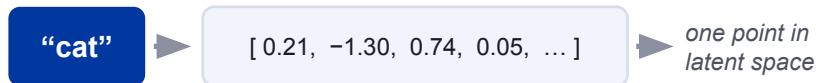


Logit lens

decode the stream at any depth

Latent space & latent vectors

Before any of the tools: the model turns every token and the running residual-stream state into a long list of numbers, i.e. a **vector** in a high-dimensional **latent space** (hundreds–thousands of dimensions). “Latent” means the meaning is never written down — it is encoded entirely in *where* the vector sits.



Position encodes meaning

- vectors that sit close together mean similar things
- directions between vectors carry meaning — the idea this whole talk turns on

LATENT SPACE · a 2-D sketch

Paris
Tokyo
Rome
cities

cat
lion
dog
animals

king → queen

man → woman

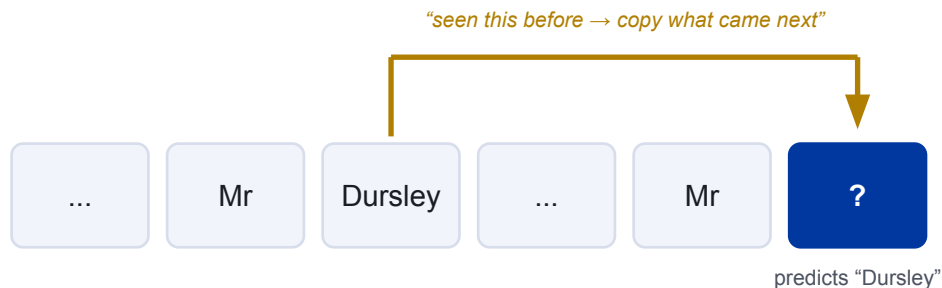
$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$

parallel arrows = one shared direction · really hundreds of dims, shown in 2-D

Everything ahead lives in this space: the residual stream is a latent vector; superposition, sparse autoencoders, and steering all operate here.

Heads route info between tokens

Attention is the only place tokens talk to each other. Each head learns a specific copy/route pattern — the cleanest example is the induction head.



Induction heads

A two-head mechanism that drives in-context learning: find a prior occurrence, copy its successor.

New vocabulary: **circuit** a named, reusable computation built from specific heads and MLPs working together.

MLP layers behave like a lookup table

Where attention moves information, the feed-forward layers store it. They act like an associative memory: a pattern in the input (the key) retrieves a stored fact (the value).

KEY (pattern in)

VALUE (fact out)

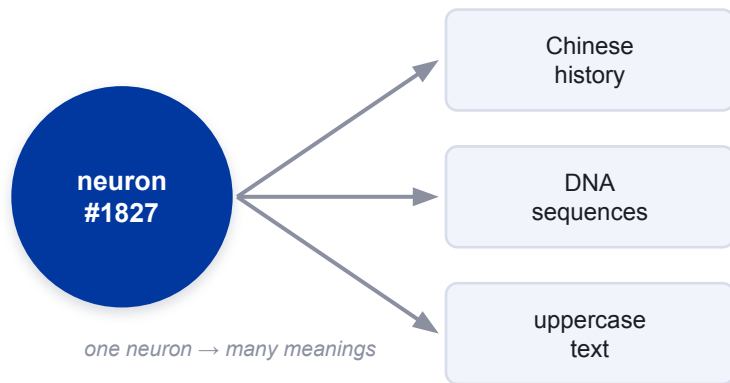


ROME / MEMIT

If facts are stored key-value pairs, you can locate and rewrite a specific one directly in the weights — strong evidence the memory model is real.

Superposition: neurons are polysemantic

Models pack far more features than they have neurons. So a single neuron fires for several unrelated concepts at once which is why reading neurons directly fails.



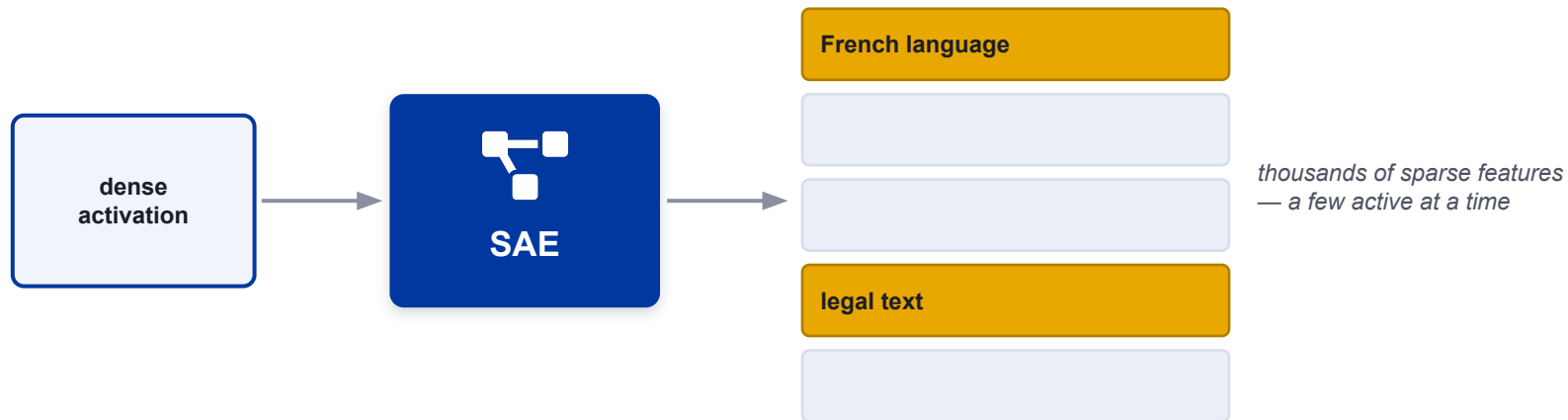
Why models do this

Most features are rare and rarely co-occur, so the model overlaps them on shared neurons trading a little interference for far more capacity.

Consequence: the unit of meaning isn't the neuron. We need a way to un-mix the overlap.

Sparse autoencoders pull concepts apart

An SAE learns to re-express a dense, polysemantic activation as a sparse combination of many monosemantic features — each one a single, interpretable concept.



Open question → are these features real properties of the model, or artifacts the SAE invented? That's what the next slide settles.

Golden Gate Claude

Find the “Golden Gate Bridge” feature. Clamp it high. The model can't stop bringing up the bridge in answers, in code, in its own self-description.

BEFORE

“What are you?”

“I’m an AI assistant made by Anthropic. How can I help you today?”

AFTER — feature clamped

“What are you?”

“I am the Golden Gate Bridge — a famous suspension bridge spanning the strait... my towers rise into the fog...”

The point isn't the gag. It's **causality** — intervening on the feature changes behavior, so features aren't mere correlations. They're real.

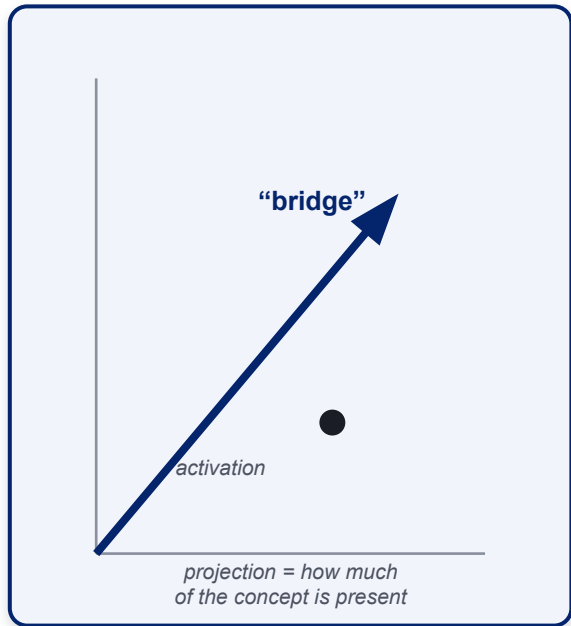
THE KEYSTONE CLAIM

Concepts are directions

in activation space.

The linear representation hypothesis, stated plainly: a concept is a direction (a vector). Its presence and strength are just the projection of the activation onto that direction.

Everything before this earned the claim. Everything after exploits it.



One claim, two superpowers

If a concept is a direction, then two operations become obvious and they define the rest of the talk.



READ the direction
= LOCATE

Measure the projection and you can find where a concept lives, and prove the model uses it.

probes · patching · circuits



ADD the direction
= STEER

Inject the vector back into the stream and you push behavior toward (or away from) the concept.

steering vectors · weight editing

Where does a concept live and can we prove it?

Locating has two halves. Finding a candidate location is easy; proving the model actually uses it is the hard, important part.
Three tools, escalating in rigor:



FIND

Linear probes

Read a concept out of activations.



PROVE

Activation patching

Causally test which sites matter.



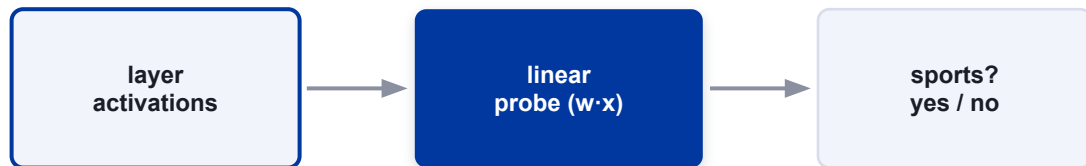
EXPLAIN

Circuits

Assemble the full mechanism.

Linear probes

Train a simple linear classifier on a layer's activations to predict a concept ("is this about sports?"). If it succeeds, the concept is linearly present at that layer.



Fast, cheap, and a great first map of where information appears across depth.

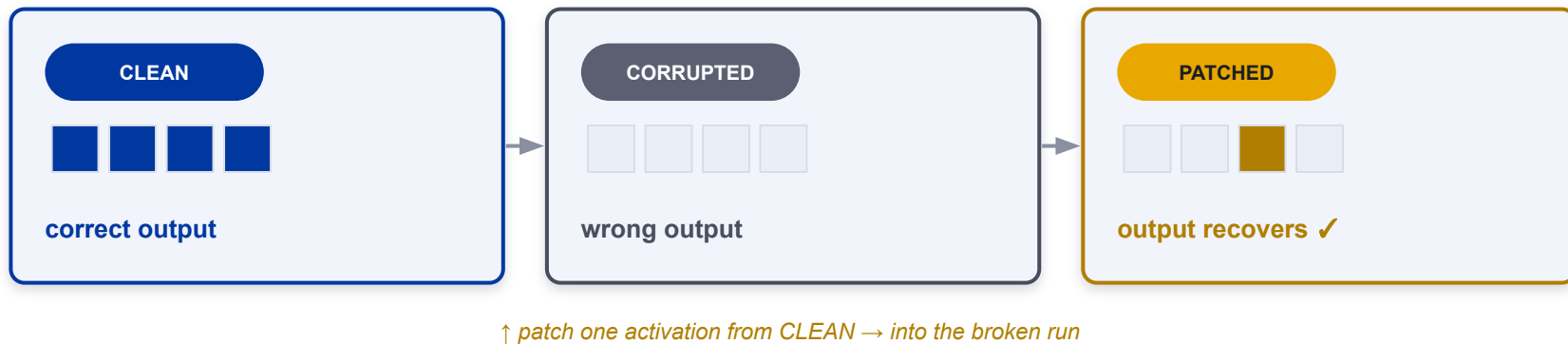


The catch

A probe shows the concept is **decodable** not that the model **uses** it. Correlation, not cause. That gap is why we need patching next.

Activation patching

Run a clean prompt and a corrupted one. Then copy a single activation from the clean run into the corrupted run. If behavior recovers, that site causally matters.



This is how you **prove** a location matters not just that a concept is readable there. Probes find; patching confirms.

Circuits: components that cooperate

Stitch located heads and MLPs into an end-to-end mechanism. The classic worked example is indirect-object identification (IOI).

“When Mary and John went to the store, John gave a drink to ____” → **Mary**



Steering vectors

Take a concept direction and add it to the residual stream at inference time. No retraining you nudge behavior live. (a.k.a. contrastive activation addition.)



Steering activations, or editing weights

Same idea move along a concept direction applied at two different places. The choice is about whether the change should last.



Steering (activations)

- Applied live at inference
- Vanishes when the run ends
- Cheap, reversible, great for probing
- Must re-apply every forward pass



Editing (weights)

- Rewrites the model itself (ROME / MEMIT)
- Permanent survives restarts
- Best for fixing a specific stored fact
- Risk of breaking unrelated knowledge

Limits & failure modes

The directions picture is powerful but leaky. Naming the cracks keeps the claims credible and motivates the frontier.



Entanglement

Directions overlap; one rarely captures a clean, isolated concept.



Off-target effects

Steering one trait drags unrelated behavior along with it.



Brittle edits

A weight edit can silently corrupt facts you didn't touch.

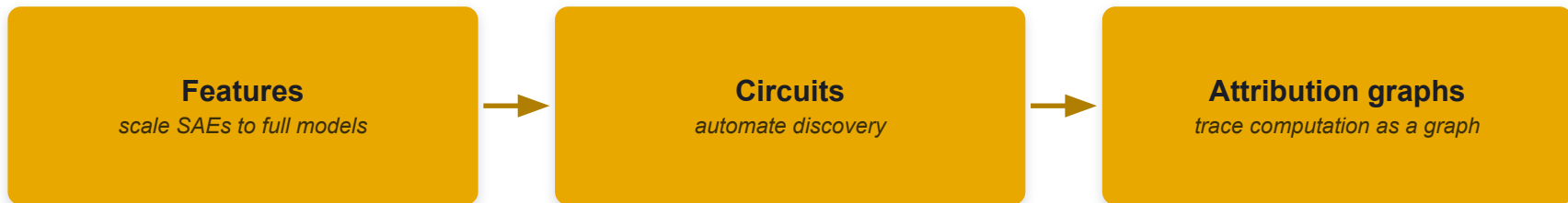


Faithfulness

A tidy explanation may not be the mechanism the model truly uses.

The research frontier

The field is scaling up: from individual features, to wired-together circuits, to whole-model attribution graphs that trace a computation end-to-end.



Still unsolved:

faithfulness

completeness

scale to frontier models

Locate · Steer · Improve

Recent surveys frame mechanistic interpretability not as observation but as an actionable pipeline. This talk covered the first two stages; the third is where the payoff lands.



Framing: Zhang et al. (2026) organize MI as Locate–Steer–Improve over interpretable objects; Rai & Yao (2024) give the task-centric taxonomy this talk's anatomy follows.

THE ONE IDEA WORTH KEEPING

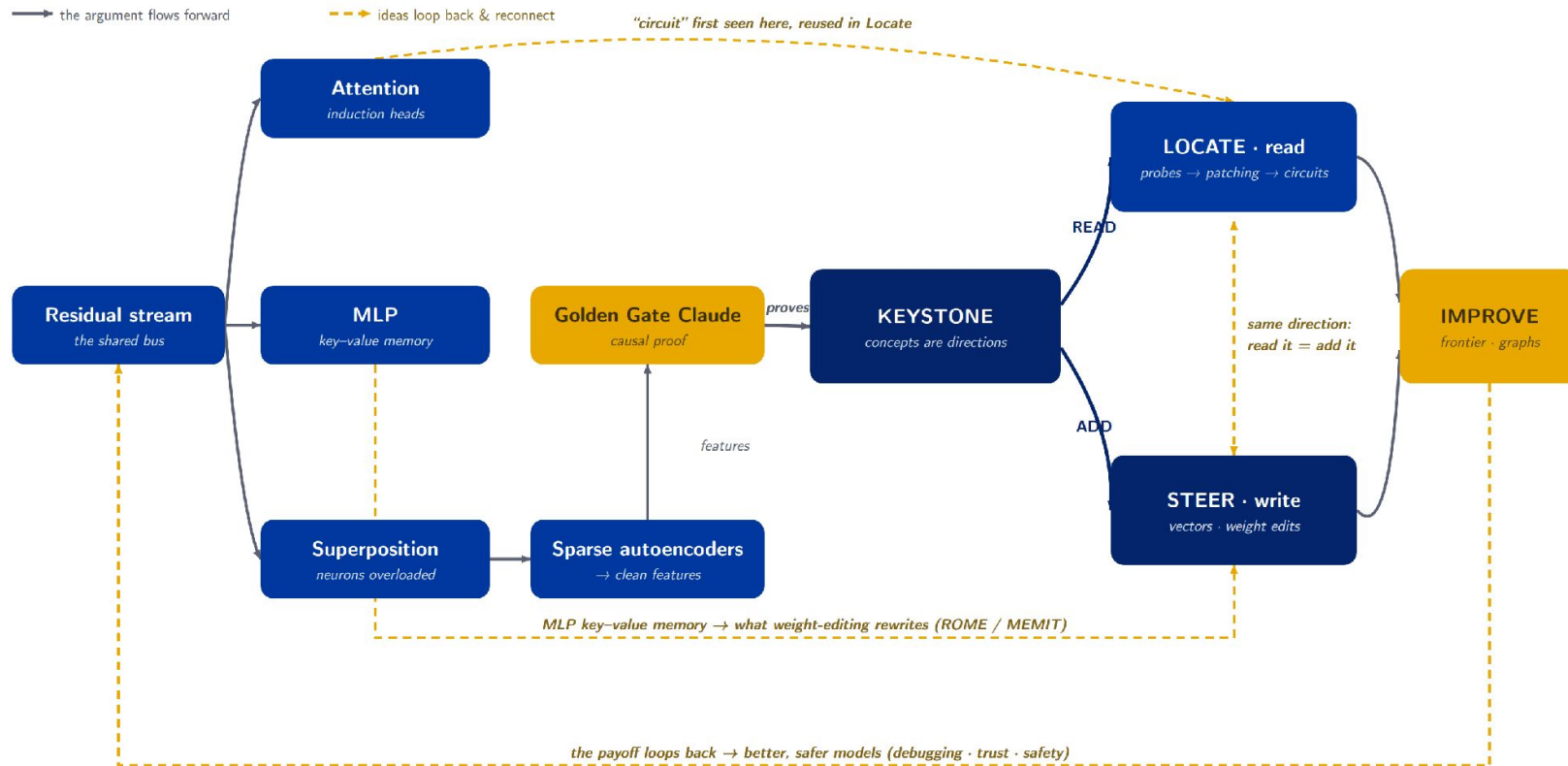
Concepts are directions which is why we can both *read* and *write* a model's cognition.



Take home

- Superposition is why neurons aren't the unit features are.
- Probes find, patching proves, circuits explain.
- The same direction you read, you can add to steer.

How it all connects



References & further reading



Framing reviews

Rai, Zhou, Feng, Saparov & Yao (2024). *A Practical Review of Mechanistic Interpretability for Transformer-Based LMs.* arXiv:2407.02646 · ICML 2025 tutorial.

Zhang et al. (2026). *Locate, Steer, and Improve: A Practical Survey of Actionable MI in LLMs.* arXiv:2601.14004.

ANATOMY

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PROOF · KEYSTONE

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LOCATE

- Meng et al. (2022). Locating & Editing Factual Associations in GPT (ROME). NeurIPS.
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- Ameisen, Lindsey et al. (2025). Circuit Tracing.
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